Chapter 85 — Performance Monitoring (Startup, Frames, Custom Metrics)

> **Part X — Performance, Observability & Quality** · Complexity: **Standard** (≥25 interview questions)
>
> **Prerequisites:** Ch 45 (startup), Ch 82 (analytics pipeline), Ch 84 (tracing), Ch 28 (privacy), Ch 44 (lifecycle). Pairs with Ch 88 (cold start), Ch 90 (jank), Ch 86 (A/B), Ch 83 (crash/vitals).
>
> **Reference surface in StreamCart:** the **field** performance signals — cold-start time, frame rendering (jank/frozen frames), screen-load latency, and custom timings (checkout duration, search latency) — captured from **real users**, reported through the analytics pipeline (Ch 82), and watched as **percentile** metrics in Android vitals and dashboards so regressions are caught.

85.1 Problem Definition and Real-World Context

You cannot improve what you don't measure, and **lab** measurements (Macrobenchmark on your dev phone, Ch 88) don't reflect what **real users on real devices/networks** experience. **Performance monitoring** is the system that captures **field** performance — **startup time**, **frame rendering** (jank/frozen frames), **screen-load latency**, and **custom timings** (checkout, search) — from production, reports them through the analytics pipeline (Ch 82), and surfaces them as **percentile** (p50/p90/p95/p99) metrics (not averages) in **Android vitals** and dashboards, so you can **detect regressions** release-over-release and by device/OS. Without it, you ship perf regressions blind and only hear about them via bad reviews.

Concretely, teams hit:

- **Lab-only measurement.** Macrobenchmark (Ch 88) is great for CI gates but measures **one device**; the field is a **long tail** of slow/old devices and bad networks. You need **real-user monitoring (RUM)**.
- **Averages hide pain.** A "good average" startup hides a terrible **p95**; perf must be tracked as **percentiles** (p50/p90/p95/p99) — the tail is where users suffer and where Android vitals thresholds bite.
- **Not measuring the right things.** The key field signals: **cold/warm start time** (Ch 45/88), **frame metrics** (jank = slow frames, **frozen frames** >700ms), **screen- load/TTID/TTFMC** (Ch 45), and **custom business timings** (checkout, search, feed load).
- **No frame monitoring.** Jank (dropped frames) is the top perceived-performance problem (Ch 90); you must capture **frame timing** in the field (`FrameMetrics`/JankStats).
- **Overhead/privacy of monitoring.** The monitoring itself must be **low-overhead** and **PII-safe** (Ch 28/82) — don't slow the app to measure it, don't log PII.
- **No regression detection/alerting.** Metrics are useless without **release-over-release** comparison and **alerting** on regressions (and gating/rollback, Ch 83/41).
- **Android vitals vs custom. Android vitals** (Play Console) gives startup/frame/ANR (Ch 83) from the field for free; **custom metrics** (Firebase Performance / your pipeline, Ch 82) add screen loads and business timings.

The architecture: capture **field performance** — **startup** (Ch 45), **frame metrics** (JankStats/`FrameMetrics`), **screen-load latency** (TTID/TTFMC, Ch 45), and **custom timings** (traces/spans, Ch 84) — from **real users**; report them through the **analytics pipeline** (Ch 82) and **Android vitals** / a perf-monitoring SDK; aggregate as **percentiles** (p50/p90/p95/p99) **by device/OS/release**; keep monitoring **low-overhead** and **PII-safe** (Ch 28); and **detect regressions** with alerting and release gating (Ch 83/ 41). This chapter is about **seeing real-user performance and catching regressions**.

Three practical use cases

1. **Startup monitoring (StreamCart).** Cold/warm start time from the field, p95 by device class, tracked release-over-release in Android vitals; a regression triggers an alert (Ch 88/83).
2. **Frame/jank monitoring.** JankStats captures janky/frozen frames per screen in production; the scroll-heavy feed (Ch 71) shows its jank rate so you know where to optimize (Ch 90).
3. **Custom timing.** A `checkout_duration` trace (Ch 84) measures how long place-order takes for real users (Ch 75), p90 by network, so slow checkouts are visible.

Why this is architectural

Performance monitoring is the **field-observability** layer for quality, composing startup (Ch 45), frame metrics, custom tracing (Ch 84), the analytics pipeline (Ch 82), privacy (Ch 28), and Android vitals/alerting (Ch 83). The decisions — measure the right field signals as percentiles, low-overhead + PII-safe capture, regression detection + gating — determine whether you can keep the app fast for real users.

85.2 Requirements Engineering

Functional requirements

ID	Requirement
F1	Capture field startup time (cold/warm), frame metrics (jank/frozen), screen-load latency (Ch 45).
F2	Capture custom business timings (checkout, search, feed) as traces/spans (Ch 84).
F3	Report via the analytics pipeline + perf SDK + Android vitals (Ch 82).
F4	Aggregate as percentiles (p50/p90/p95/p99) by device/OS/release — not averages.
F5	Keep monitoring low-overhead and PII-safe (Ch 28).
F6	Detect regressions release-over-release; alert and gate/rollback (Ch 83/41).

Non-functional requirements

Attribute	Policy
Fidelity	Real-user (field), not lab; percentiles capture the tail.
Overhead	Low-overhead capture; doesn't slow the app.
Privacy	PII-safe; consent regime (Ch 28/82).
Actionability	By device/OS/release; regression alerting (Ch 83/41).
Coverage	Startup + frames + screen loads + custom timings.

Scope

- **In scope:** field startup (Ch 45/88), frame metrics (JankStats/ `FrameMetrics`, jank/ frozen), screen-load latency (TTID/TTFMC, `reportFullyDrawn`), custom timings (traces/spans, Ch 84), reporting via analytics (Ch 82) + perf SDK + Android vitals, percentile aggregation by device/OS/release, low-overhead + PII-safe (Ch 28), regression detection/alerting/gating (Ch 83/41), RUM vs lab (Ch 88).
- **Out of scope:** startup optimization (Ch 45/88), jank fixing (Ch 90), analytics pipeline (Ch 82), tracing (Ch 84), crash/ANR (Ch 83), Macrobenchmark detail (Ch 88).

Assumptions & constraints

- Analytics pipeline (Ch 82) + a perf-monitoring SDK (Firebase Performance / custom) + **Android vitals** (Play Console). JankStats/ `FrameMetrics` for frames; `reportFullyDrawn` / traces for screen loads (Ch 45). `minSdk 24`. Monitoring is **low-overhead** and **PII- safe**; metrics are **percentiles** by device/OS/release.

85.3 System Design Fundamentals

- **Field (RUM) vs lab (Ch 88). Lab** tools (Macrobenchmark, Ch 88) measure **one controlled device** — essential for **CI gates** and reproducibility, but they don't see the **field's long tail** (old/slow devices, bad networks, memory pressure). **Field / Real-User Monitoring (RUM)** captures what **real users** experience. You need **both**: lab for gating, field for truth. This chapter is **field** monitoring.
- **Measure in percentiles, not averages (the key statistical point).** An **average** hides the tail — a 1s mean startup can hide a 5s **p95**. Track **p50/p90/p95/p99** (and max): the **tail** (p95/p99) is where users suffer, where bad reviews come from, and where **Android vitals thresholds** apply. Always report and alert on **percentiles** segmented by **device class/OS/release**.
- **The field signals to capture:**
  - **Startup time (Ch 45/88): cold** (process start → first frame/fully-drawn) and **warm**; TTID (time to initial display) and TTFMC (`reportFullyDrawn`). Android vitals reports field startup; a perf SDK can too.
  - **Frame metrics (jank): janky frames** (slow frames missing the frame deadline) and **frozen frames** (>700ms) — the top perceived-perf signal (Ch 90). Capture via **JankStats** (Jetpack) or `Window.addOnFrameMetricsAvailableListener` — per screen, in the field.
  - **Screen-load latency:** TTID/TTFMC per screen (Ch 45) — how long until a screen is usable; custom traces around navigation.

    - **Custom business timings: checkout duration** (Ch 75), **search latency** (Ch 9), **feed load** (Ch 71) — measured as **traces/spans** (Ch 84) so you track what matters to the product, not just system metrics.
- **Report via the analytics pipeline + perf SDK + Android vitals (Ch 82).** Performance metrics flow through the **same reliable, batched, consented, PII-safe pipeline** as analytics (Ch 82) — durably buffered, batched, redacted. **Android vitals** (Play Console) gives startup/frames/ANR (Ch 83) for free from the field; a **perf SDK** (Firebase Performance) or your pipeline adds **custom** screen loads and business timings. Correlate with **tracing** (Ch 84) where a specific trace is slow.
- **Low-overhead, PII-safe capture (Ch 28).** The monitoring must **not slow the app** — frame-metrics listeners are cheap, custom traces are lightweight, and everything is **batched** (Ch 82). Metrics are **PII-safe**: they measure timing, not personal data; log device/OS/screen, not user content (Ch 28). Respect **consent** (Ch 82).
- **Regression detection + alerting + gating (Ch 83/41).** Metrics are only useful if you **compare release-over-release** (did this release make startup/jank worse?), **by device/ OS**, and **alert** on regressions — then **gate/roll back** a bad release (Ch 41) like crashes (Ch 83). Android vitals flags bad behavior that affects **store ranking**. Set **perf budgets/SLOs** (e.g., p95 cold start < X) and alert on breach.
- **Sampling (cost).** High-volume per-frame data is **sampled/aggregated** (don't export every frame); startup/custom timings may be sampled too — balance fidelity vs cost (Ch 82).
- **Correlate perf with context.** Attach device class, network type (Ch 68), app version, and a **correlation id** (Ch 84) so a slow timing can be **investigated** (which trace, which device). Perf without context isn't actionable.

Core insight: **monitor real-user (field) performance — not just lab (Ch 88) — capturing the signals that matter: cold/warm startup (Ch 45), frame metrics (janky and >700ms frozen frames via JankStats), screen-load latency (TTID/TTFMC), and custom business timings (checkout, search, feed) as traces/spans (Ch 84); report them through the same reliable, batched, consented, PII-safe pipeline as analytics plus Android vitals (Ch 82), aggregated as percentiles (p50/p90/p95/p99 — never averages, because the tail is where users suffer) segmented by device/OS/release; keep capture low-overhead and PII-safe (Ch 28); and detect regressions release-over-release with alerting, SLOs/budgets, and gating/rollback (Ch 83/41) — correlating slow metrics with trace/device context (Ch 84) so they're actionable.**

85.4 High-Level Architecture



Components & ownership

- **Capture** field startup/frames/screen-loads/custom-timings (low-overhead, Ch 45/84).
- **Report** via the analytics pipeline + perf SDK + **Android vitals** (Ch 82).
- **Aggregate as percentiles** by device/OS/release; **detect regressions** → alert/gate (Ch 83/41).
- **Correlate** with context (Ch 84); **PII-safe** + consented (Ch 28).

Direction: field capture → pipeline (Ch 82) + vitals → percentile aggregation by device/release → regression alerting/gating; correlated + PII-safe.

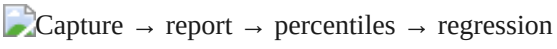
85.5 Detailed Component Design



Component	Responsibility
`PerfMonitor`	capture startup/frames/custom timings (low-overhead, Ch 45/84)
`PerfMetric`	a timing + context (device/OS/release/correlation, Ch 84)
`PerfReporter`	report via the analytics pipeline, PII-safe (Ch 82/28)
`RegressionWatcher`	percentile aggregation + regression alerting (Ch 83/41)

85.6 End-to-End Data Flow

Capture → report → percentiles → regression



Percentiles vs average



85.7 Complete Kotlin Implementation

Package: `com.streamcart.perf`.

85.7.1 Frame metrics via JankStats (jank/frozen frames)

```
@Composable
fun MonitoredScreen(screen: String, content: @Composable () -> Unit) {
    val window = (LocalContext.current as Activity).window
    DisposableEffect(window) {
        val jankStats = JankStats.createAndTrack(window) { frameData ->   // JankStats (Jetpack)
            if (frameData.isJank) perfReporter.report(PerfMetric(
                name = "frame_jank", durationMs = frameData.frameDurationUiNanos / 1_000_000,
                type = if (frameData.frameDurationUiNanos > 700_000_000) "frozen" else "janky",
// >700ms = frozen
                screen = screen, /* + device/OS/release context */))
        }
        onDispose { jankStats.isTrackingEnabled = false }   // low-overhead; scoped to screen
(Ch 44)
    }
    content()
}
```

85.7.2 Startup monitoring (field, Ch 45)

```
// Android vitals reports field startup for free. For custom: reportFullyDrawn marks TTFMC (Ch
45).
class StartupMonitor @Inject constructor(private val reporter: PerfReporter) {
    fun onFullyDrawn(activity: ComponentActivity, startNanos: Long) {
        activity.reportFullyDrawn()   // marks TTFMC for vitals/traces (Ch 45)
        reporter.report(PerfMetric(name = "cold_start_ttfmc",
            durationMs = (System.nanoTime() - startNanos) / 1_000_000, type = "startup"))   //
custom field metric
    }
    // Perf SDK (Firebase Performance) auto-captures _app_start trace; add custom for screen
loads.
}
```

85.7.3 Custom business timings as traces (Ch 84)

```
@Singleton
class PerfMonitor @Inject constructor(private val reporter: PerfReporter, private val
correlation: CorrelationManager) {  // Ch 84
    /** Measure a business operation (checkout/search/feed) end-to-end. */
    suspend fun <T> trace(name: String, block: suspend () -> T): T {
        val start = System.nanoTime()
        return try { block() } finally {
            reporter.report(PerfMetric(name = name, durationMs = (System.nanoTime() - start) /
1_000_000,
                type = "custom", correlationId = correlation.currentCorrelationId()))   //
correlate (Ch 84)
        }
    }
}
// Usage: perfMonitor.trace("checkout_duration") { placeOrderUseCase.place(session) }  (Ch 75)
//        perfMonitor.trace("search_latency") { searchRepo.search(query) }             (Ch 9)
```

85.7.4 PII-safe reporting via the analytics pipeline (Ch 82/28)

```
@Singleton
class PerfReporter @Inject constructor(private val analytics: Analytics, private val device:
DeviceContext) {  // Ch 82
    fun report(metric: PerfMetric) {
        analytics.track(AnalyticsEvent.Perf(                        // same reliable, batched,
consented pipeline (Ch 82)
            name = metric.name, durationMs = metric.durationMs,
            params = mapOf(                                          // CONTEXT - no PII (Ch
28)
                "type" to metric.type, "device_class" to device.deviceClass, "os" to
device.osVersion,
                "release" to device.appVersion, "network" to device.networkType,
// Ch 68
                "correlation_id" to metric.correlationId, "screen" to metric.screen)))
    }
    // No user content/PII; timing + device/OS/release/screen context only.
}
```

85.7.5 Percentiles + regression watch (dashboard/backend side)

```
// Aggregation is backend/dashboard-side (Android vitals / Firebase / your pipeline):
//  - compute p50/p90/p95/p99 by device_class x os x release (NOT averages)
//  - compare release-over-release -> regression -> alert + gate/rollback (Ch 83/41)
//  - SLOs/budgets: e.g., p95 cold_start < 2s on mid-tier; p95 frame jank rate < X%
```

85.7.6 Module structure

```
:core:perf  PerfMonitor (frames/startup/custom), PerfMetric, PerfReporter (via analytics, Ch 82)
# Startup -> Ch 45; frames -> JankStats; custom -> traces (Ch 84); vitals -> Play Console.
# Aggregation/regression/alerting -> backend/vitals (Ch 83/41). PII-safe -> Ch 28.
```

85.8 Concurrency and Race Conditions

Scenario	Risk	Handling
Monitoring slows the app	perf paradox	low-overhead capture (JankStats/listeners); batch (Ch 82)
Averages reported	tail hidden	percentiles (p50/p90/p95/p99)
No device segmentation	misleading	segment by device class/OS/release
Lab-only	field tail missed	RUM (field) + lab (Ch 88)
Per-frame flood	cost	sample/aggregate frames (Ch 82)
PII in metrics	leak	timing + context only, no content (Ch 28)
Regression unnoticed	shipped slow	release-over-release + alerting (Ch 83/41)

- **Low-overhead is essential** (the monitoring must not degrade what it measures): frame listeners/JankStats are cheap, custom traces are lightweight, and reporting is **batched** through the analytics pipeline (Ch 82) — never block the UI or spam the network to measure.
- **Percentiles by segment** are the only honest view: an average or an un-segmented number hides the **tail** and the **slow-device** population where perf actually hurts.
- **Field + lab together:** RUM shows the real distribution; Macrobenchmark (Ch 88) gives a reproducible CI gate — neither alone suffices.

85.9 Failure Scenarios and Recovery

Failure	Detection	Response	Outcome
Perf regression shipped	p95 spike	release-over-release alerting + rollback (Ch 83/41)	caught
"Average is fine" but users complain	reviews	track percentiles (tail)	visible
Jank unmeasured	user reports	JankStats frame metrics	measured
Lab passed, field slow	field data	RUM captures the tail (Ch 88)	surfaced
Monitoring slows app	own metrics	low-overhead + batched (Ch 82)	negligible
PII in perf data	privacy audit	context-only, PII-safe (Ch 28)	safe
Not actionable	can't fix	correlate device/network/trace (Ch 84/68)	actionable


85.9 Failure Scenarios and Recovery

> **Monitor real users in percentiles, by device/release, and alert on regressions.** The field's tail (p95/p99 on slow devices/networks) is where performance actually hurts and where Android vitals thresholds bite — averages and lab-only measurement miss it. Capture startup/frames/screen-loads/custom-timings low-overhead through the analytics pipeline, and gate releases on perf like you do on crashes (Ch 83).

85.10 Scaling the Implementation

- **Level 1.** Android vitals (startup/frames/ANR) from the field; percentile awareness.
- **Level 2.** JankStats frame monitoring per screen; custom timings (traces, Ch 84) via the analytics pipeline (Ch 82); PII-safe + consent (Ch 28).
- **Level 3 — production.** Percentile dashboards by device/OS/release; regression alerting + gating/rollback (Ch 83/41); SLOs/budgets; context correlation (Ch 84/68).
- **Level 4 — large.** Per-screen/flow perf budgets; adaptive sampling; perf experiments (Ch 86); device-tiered targets; trace-based deep investigation (Ch 84).
- **Level 5 — enterprise.** Perf SLOs with error budgets, automated regression gating in CI (lab, Ch 88) + field alerting, perf governance, cross-platform parity, long-tail device programs.

Avoid overengineering: Android vitals + JankStats + a few custom timings through the analytics pipeline, tracked as percentiles with regression alerting, covers most needs. Don't measure averages; don't add heavy monitoring; don't rely on lab alone.

85.11 Alternative Approaches and Trade-offs

Dimension	Option A	Option B	When
Where	Field (RUM)	lab (Macrobenchmark, Ch 88)	Both: field for truth, lab for CI gate
Stat	Percentiles (p50/p90/p95/p99)	averages	Percentiles always (tail)
Segmentation	By device/OS/release	aggregate	Segment (slow-device tail)
Frames	JankStats/ `FrameMetrics`	none	JankStats (jank is top signal)
Custom timings	Traces/spans (Ch 84)	nothing	Traces (business metrics)
Reporting	Analytics pipeline + vitals (Ch 82)	ad-hoc	Pipeline (reliable, consented, PII-safe)
Action	Regression alert + gate (Ch 83/41)	dashboards only	Alert+gate (catch regressions)

Field (RUM) vs lab (Macrobenchmark, Ch 88): lab measures one controlled device reproducibly — ideal for **CI gating** — but can't see the **field's long tail** (old devices, bad networks, memory pressure). **Field RUM** captures the real distribution users experience. Use **both**: Macrobenchmark to **gate** regressions in CI (Ch 88), RUM to **see truth** and alert on field regressions.

Percentiles vs averages: an **average** collapses the distribution and **hides the tail** — a good mean can mask a terrible p95 that drives bad reviews and trips Android vitals thresholds. **Percentiles (p95/p99)** segmented by device/OS/release show where users actually suffer. Always measure and alert on percentiles, never averages.

Analytics pipeline vs ad-hoc reporting: sending perf metrics ad-hoc (inline, unbatched, unconsented) repeats the analytics mistakes (Ch 82) — lost/expensive/PII-risky. Routing perf through the **same reliable, batched, consented, PII-safe pipeline** (Ch 82) gives durable, efficient, compliant reporting. Reuse the pipeline.

85.12 Security & Privacy Considerations

- **PII-safe metrics (Ch 28):** performance data is **timing + device/OS/screen context**, **not** personal data — never attach user content, PII, or secrets. Screen names should be non-sensitive labels (Ch 28).
- **Consent (Ch 28/82):** perf monitoring follows the same **consent** regime as analytics; if the user opted out, limit to essential/anonymous.
- **Correlation ids opaque (Ch 84):** link slow metrics to traces via **opaque** ids, not PII.
- **No sensitive screens in traces:** a custom trace around a payment screen (Ch 75) measures **timing**, not content — never capture screen content.
- **Secure, access-controlled dashboards:** perf data is internal; access-controlled, retention-limited (Ch 28).
- Client vs backend: the **client** captures low-overhead PII-safe timings and reports via the pipeline; the **backend/vitals** aggregates percentiles, detects regressions, and access-controls — neither stores PII in perf data.

85.13 Performance and Resource Optimization

- **Low-overhead capture** — JankStats/frame listeners are cheap; custom traces are lightweight; **batch** reporting (Ch 82). Monitoring must not degrade perf.
- **Sample high-volume** — per-frame data sampled/aggregated; startup/custom timings sampled as needed (Ch 82).
- **Scope frame tracking to visible screens** (Ch 44) — don't track off-screen.
- **Report via the batched pipeline** (Ch 82) — no inline/synchronous sends.
- **Minimal context** — device/OS/release/screen (cheap), not heavy payloads.
- **Profiling:** monitoring overhead (should be negligible), report volume, percentile computation cost (backend), sampling rate.

85.14 Testing Strategy

Capture logic and PII-safety are testable; percentiles/regression validated with field data + staged rollouts.

```
class PerfReporterTest {
    @Test fun `reports timing with context, no PII`() {
        reporter.report(PerfMetric("checkout_duration", 1200, "custom", correlationId = "c1"))
        val e = analytics.lastPerfEvent()
        assertEquals(1200, e.durationMs); assertTrue(e.params.containsKey("device_class"))
        assertFalse(e.params.values.any { it.toString().contains("@") })   // no PII (Ch 28)
    }
}

class PerfMonitorTest {
    @Test fun `trace measures duration and correlates`() = runTest {
        perfMonitor.trace("search_latency") { delay(100) }
        assertTrue(analytics.lastPerfEvent().durationMs >= 100)
        assertEquals("c1", analytics.lastPerfEvent().params["correlation_id"])   // Ch 84
    }
}

class JankTest {
    @Test fun `classifies frozen vs janky frames`() { /* >700ms -> frozen */ }
}
// Macrobenchmark (Ch 88) for lab gating; staged rollout to watch field percentiles before full
release.
```

What to test: custom-trace timing + correlation (Ch 84); PII-safe reporting (context only, Ch 28); frame classification (janky/frozen); startup metric (`reportFullyDrawn`); reporting via the pipeline (batched, Ch 82). Lab gating via Macrobenchmark (Ch 88); field percentiles via staged rollouts.

85.15 Production Observability

- **Metrics (the point): cold/warm startup** p50/p90/p95/p99 by device class/OS/release (Android vitals + custom); **frame jank rate** + **frozen-frame rate** per screen; **screen- load latency** (TTID/TTFMC); **custom business timings** (checkout/search/feed) percentiles; monitoring overhead (negligible); report delivery (Ch 82).
- **Logs:** perf metrics are events in the pipeline (Ch 82) — **no PII** (Ch 28).
- **Debugging:** regression → release-over-release percentile spike (which device/OS); jank → frame metrics per screen (Ch 90); slow startup → cold-start trace (Ch 88); slow checkout → custom trace + correlation (Ch 84/75); "average fine but complaints" → look at p95/p99.
- **Alerting:** perf **regression** on a new release (p95 breach → gate/rollback, Ch 83/41); SLO/budget breach (e.g., p95 cold start); jank-rate spike.
- **Dashboards:** startup/jank/screen-load/custom-timing **percentiles** by device/OS/release; release-over-release trends; the **tail**.

85.16 System Design Interview Preparation

> ≥25 questions.

Fundamentals

1. *Field vs lab?* RUM (real users) vs Macrobenchmark (one device, CI gate, Ch 88) — use both.
2. *Why field?* The long tail (slow devices/networks) isn't in the lab.
3. *Percentiles vs averages?* Averages hide the tail; track p50/p90/p95/p99.
4. *Why the tail matters?* p95/p99 is where users suffer + vitals thresholds apply.
5. *Segment by?* Device class, OS, release.

Signals 6. *Startup?* Cold/warm, TTID/TTFMC (Ch 45); Android vitals + custom. 7. *Frames?* Janky + frozen (>700ms) via JankStats/`FrameMetrics`. 8. *Screen load?* TTID/TTFMC per screen. 9. *Custom timings?* Business ops (checkout/search/feed) as traces (Ch 84). 10. *Why custom?* Track what matters to the product.

Reporting 11. *How report?* Analytics pipeline (Ch 82) + perf SDK + Android vitals. 12. *Why the pipeline?* Reliable, batched, consented, PII-safe (Ch 82). 13. *Android vitals?* Field startup/frames/ANR for free. 14. *Correlate?* Device/network/version + correlation id (Ch 84).

Overhead/privacy 15. *Overhead?* Low — cheap listeners + batched; must not slow the app. 16. *Sampling?* High-volume (frames) sampled/aggregated. 17. *PII?* Timing + context only (Ch 28); consent (Ch 82).

Regression 18. *Detect?* Release-over-release percentiles by device/OS. 19. *Act?* Alert + gate/rollback (Ch 83/41); SLOs/budgets. 20. *Vitals ranking?* Bad perf affects store visibility.

Lab/field 21. *CI gate?* Macrobenchmark (Ch 88). 22. *Field truth?* RUM.

Testing/Production 23. *Test capture?* Trace timing + PII-safe reporting. 24. *"Average fine" complaints?* Look at p95/p99. 25. *Jank in prod?* Frame metrics per screen. 26. *Key metric?* Startup/jank percentiles by device/release.

Design-change challenges

1. **Measure real-user startup.** Android vitals + custom cold-start metric, p95 by device (Ch 45/88).
2. **Find janky screens.** JankStats frame metrics per screen (Ch 90).
3. **Average looks fine but users complain.** Track percentiles (p95/p99) by device.
4. **Catch a perf regression before full rollout.** Staged rollout + release-over-release alerting (Ch 83/41).
5. **Measure checkout latency for real users.** Custom trace + correlation (Ch 84/75).

Mock walkthrough (abridged)

Clarify (what to measure, device targets, regression gating?) → field RUM + lab gate (Ch 88) → capture startup (Ch 45) + frames (JankStats) + screen loads + custom timings (traces, Ch 84) → report via analytics pipeline + vitals (Ch 82), PII-safe (Ch 28) → aggregate percentiles by device/OS/release (not averages) → correlate context (Ch 84/68) → regression alerting + SLOs + gate/rollback (Ch 83/41) → low-overhead → observability (tail percentiles).

85.17 Practical Exercises

1. **Beginner.** Capture frame metrics per screen with JankStats and report janky/frozen frames via the analytics pipeline (Ch 82). *Eval:* jank measured; PII-safe.
2. **Intermediate.** Add a custom `checkout_duration` trace (Ch 84/75) and a cold-start metric (`reportFullyDrawn`, Ch 45), reported with device/OS/release context. *Eval:* timings captured + correlated.
3. **Advanced.** Aggregate as percentiles by device class and add release-over-release regression alerting (Ch 83/41). *Eval:* p95 by device; regression alerts.
4. **Production debugging.** Investigate "average startup fine but complaints" by looking at p95/p99 on slow devices; find a janky screen. *Eval:* tail surfaced; screen identified.
5. **System design challenge.** Combine Macrobenchmark CI gating (Ch 88) with field RUM alerting and perf SLOs/budgets. *Eval:* lab gate + field truth; budget alerts.

(Solutions withheld unless requested.)

85.18 Chapter Summary & Reference Sheet

Key decisions: monitor **real-user (field) performance** — not just lab (Macrobenchmark, Ch 88) — capturing the signals that matter: **cold/warm startup** (TTID/TTFMC, Ch 45), **frame metrics** (**janky** and **>700ms frozen** frames via **JankStats**/ `FrameMetrics`), **screen- load latency** per screen, and **custom business timings** (checkout, search, feed) as **traces/spans** (Ch 84); **report them through the same reliable, batched, consented, PII-safe analytics pipeline** as events plus **Android vitals** (Ch 82); **aggregate as percentiles** (**p50/p90/p95/p99 — never averages**, because the **tail** on slow devices/networks is where users suffer and where vitals thresholds apply) **segmented by device class/OS/release**; keep capture **low-overhead** and **PII-safe** (timing + context only, consented, Ch 28); **correlate** slow metrics with device/network (Ch 68) and trace/correlation ids (Ch 84) so they're **actionable**; and **detect regressions release-over-release** with **alerting, SLOs/budgets, and gating/rollback** (Ch 83/41) — using **both** field RUM (truth) and lab Macrobenchmark (CI gate, Ch 88).

Concepts: field RUM vs lab (Ch 88), percentiles (p95/p99) vs averages, startup (Ch 45), frame metrics (jank/frozen, JankStats), screen-load latency, custom timings (traces, Ch 84), reporting via the analytics pipeline + Android vitals (Ch 82), device/OS/release segmentation, low-overhead + PII-safe (Ch 28), regression alerting/gating (Ch 83/41).

Common mistakes: lab-only (missing the field tail); tracking averages (hiding p95/p99); no segmentation (slow-device population hidden); not measuring frames/jank; heavy monitoring that slows the app; PII in perf data; ad-hoc reporting (not the pipeline); no regression detection/ alerting (shipping slow blind).

Checklist: ☐ field RUM + lab gate (Ch 88) ☐ capture startup (Ch 45) + frames (JankStats, janky/frozen) + screen loads + custom timings (traces, Ch 84) ☐ report via analytics pipeline

- Android vitals (Ch 82) ☐ percentiles (p50/p90/p95/p99), not averages ☐ segment by device/OS/ release ☐ low-overhead + PII-safe + consent (Ch 28) ☐ correlate device/network/trace (Ch 84/

68. ☐ regression alerting + SLOs/budgets + gate/rollback (Ch 83/41) ☐ sample high-volume frames ☐ tail (p95/p99) dashboards.

Key APIs: `JankStats` (Jetpack frame metrics) / `Window.addOnFrameMetricsAvailableListener`, `ComponentActivity.reportFullyDrawn` (TTFMC, Ch 45), Firebase Performance / custom traces (Ch 84), analytics pipeline (Ch 82), Android vitals (Play Console), Macrobenchmark (lab gate, Ch 88), percentile aggregation (backend/vitals), regression alerting (Ch 83/41).

References:

- Android vitals: https://developer.android.com/topic/performance/vitals
- JankStats: https://developer.android.com/topic/performance/jankstats
- Firebase Performance Monitoring: https://firebase.google.com/docs/perf-mon
- Measure app startup (field): https://developer.android.com/topic/performance/vitals/launch-time
- (composes) Ch 45 Startup, Ch 82 Analytics, Ch 84 Tracing, Ch 88 Cold Start/Macrobenchmark, Ch 90 Jank, Ch 83 Crash/Vitals, Ch 41 Release/Rollback, Ch 28 Privacy

Chapter review (Phase 4)

- **Completeness:** all 18 sections; architecture/class/sequence/percentiles/failure diagrams; JankStats + startup + custom-trace + PII-safe reporter code; concurrency (low-overhead, percentiles-by-segment, field+lab); failure (percentiles+field+regression principle); 5 levels; field-vs-lab, percentiles-vs-averages, pipeline trade-offs; security (PII-safe, consent, opaque correlation); perf (low-overhead, sample, batched); reporter + trace + jank tests; observability (tail percentiles by device/release); 26 Q&A + 5 challenges. ✅
- **Expand later:** startup/cold start (Ch 45/88), jank (Ch 90), analytics (Ch 82), tracing (Ch 84), crash/vitals (Ch 83).
- **Interview coverage:** exceeds ≥25 bar.